

Paper 24.25 - KnightForge Toolkit Supplement

CQE-paper-24.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-24.25.md

Paper 24.25 - KnightForge Toolkit Supplement

Purpose

This supplement shows how to reproduce the KnightForge receipt. The proof is in Paper 24 and its formal verifier; this file holds the tool and analog handling.

Digital Route

Run:

```
`python production/formal-papers/CQE-paper-24/verify_knightforge_ca.py`
```

The expected status is ``pass_with_open_obligations``. The verifier checks centroid annealing, the ``2 + 6`` sector split, finite greedy knight placement, per-row L-conjugate closure, and the N-dimensional frame-operator boundary.

Analog Route

Draw a numbered board. Sweep cells in order. At each cell, mark the two opposed knight-approach families as ``L`` and ``R``; mark ``C = 1`` if the cell is occupied and ``C = 0`` if the cell is rejected. Draw the anneal route by swapping the three positions until ``L = R``. The analog row passes if the same attractor and step count are recovered.

Hidden-Guess Diagnostic

Training mode should hide the label until after the reviewer chooses one:

- occupied row,
- rejected row,
- attacking invalid row,
- L-conjugate closure,
- sector split,
- frame operator,
- invalid solved-game promotion.

The answer key distinguishes local CA closure from game-theoretic closure.